

KOGI PLATFORM

kogi-game

Game System & Mechanism Engine

System Design Document

v1.0 · Kogi Platform · March 2026

KGAM · kogi-game · kogi-engine · MatchEngine · AllocEngine · IncentiveEngine

*Resource · Capital · Labor · Exchange · Marketplace · Portfolio · Assets
Bids · Offers · Deals · Requests · Proposals · Contracts · Gigs · Tasks
Matching · Allocation · Incentive Mechanisms · Recommendations · Personalization*

Kogi Team · Independent Worker Operating System · Confidential

Table of Contents

1. OVERVIEW & PHILOSOPHY

1.1 What is kogi-game?

kogi-game is the mechanism design and game engine of the Kogi Platform — the mathematical and computational substrate that governs how resources, capital, labor, portfolio components, and financial instruments are discovered, matched, priced, allocated, exchanged, and incentivized across the platform.

Every transaction on the Kogi Platform is, at its core, an economic game: a buyer and seller seeking mutually beneficial terms, a worker and opportunity seeking optimal fit, an investor and portfolio seeking the best risk-adjusted return, a collective seeking a fair distribution rule. kogi-game formalizes these games — specifying the rules, the scoring functions, the information revelation protocols, and the equilibrium-seeking mechanisms — so that outcomes are efficient, fair, and incentive-compatible for all participants.

Design Principle: Mechanism design is the engineering of economic games. Where economics asks 'what outcomes do given rules produce?', mechanism design asks 'what rules produce the outcomes we want?' kogi-game answers that question for every interaction on the platform — from a single gig bid to a multi-million dollar equity round — designing rules that align individual incentives with platform-wide efficiency.

1.2 The Four-Layer Architecture

kogi-game operates as four interdependent layers, each building on the layer below:

Layer	Name	Responsibility	Implementation
L4 (Top)	Incentive Layer	Reputation, rewards, stakes, and long-run behavior shaping	IncentiveEngine — Go service + kogi-engine signals
L3	Allocation Layer	Deciding who gets what, when, at what price, in what order	AllocEngine — Rust core + Go service
L2	Matching Layer	Finding optimal pairings across all entity types	MatchEngine — Scala (kogi-engine) + Go service
L1 (Base)	Signal Layer	Collecting, normalizing, and distributing the data that all higher layers depend on	TelemetryEngine + AnalyticsEngine (kogi-engine)

1.3 Entities in the Game

Every entity in the Kogi Platform is a player in one or more games. The game system categorizes all participants and resources:

Entity Class	Platform Types	Game Role
Labor	Independent workers, contractors, freelancers, gig workers, team members	Sellers of time, skills, and deliverables; bidders in RFP auctions
Capital	Investors, donors, lenders, campaign backers, grant providers	Allocators of financial resources; participants in investment matching
Resources	Compute, licenses, materials, equipment, tools, data, templates	Supply-side of resource allocation games; inputs to production
Portfolio Components	Programs, Projects, Assets, Artifacts, Resources (Portfolio System kinds)	Objects of investment, trade, and collaboration; generate signals
Financial Instruments	Equity, revenue-share notes, bonds, tokens, derivatives	Traded in kogi-exchange financial instruments market
Demand-side entities	Buyers, hirers, organizations, cooperatives, teams	Create demand signals; fund escrow; initiate matching
Transactional entities	Bids, Offers, Deals, Requests, Proposals, Contracts, Gigs, Tasks	Game moves — the atomic actions through which players interact
Campaigns	Fundraise, equity, group-buy, resource, bond, pre-sale	Coordination games — multiple buyers coordinating on a shared purchase or investment
Collectives/Cooperatives	Worker collectives, DAOs, federated entities	Cooperative games — designing fair distribution rules within groups

1.4 Service Architecture

Service	Code	Technology
kogi-game	KGAM	Go (primary service) + Rust (core mechanisms)
kogi-engine MatchEngine	KMTCH	Scala 3, gRPC port 9100
kogi-engine AllocEngine	KALLOC	Scala 3, embedded in kogi-engine
kogi-engine IncentiveEngine	KINC	Scala 3, embedded in kogi-engine
kogi-engine RecommendationEngine	KREC	Scala 3, embedded in kogi-engine
kogi-engine PersonalizationEngine	KPERS	Scala 3, embedded in kogi-engine

kogi-engine AnalyticsEngine	KANL	Scala 3, embedded in kogi-engine
kogi-engine RiskEngine	KRSK	Scala 3, embedded in kogi-engine
kogi-engine GraphEngine	KGRPH	Scala 3, embedded in kogi-engine

2. MECHANISM DESIGN FOUNDATIONS

2.1 Core Principles

kogi-game's mechanism design is grounded in the following economic design principles, applied pragmatically to the independent worker economy context:

Principle	Application in kogi-game
Incentive Compatibility	Mechanisms are designed so that the dominant strategy for every participant is to reveal their true preferences — honest bidding, accurate skill declaration, genuine availability — because truthful behavior maximizes their payoff.
Individual Rationality	No participant is made worse off by participating. Every mechanism guarantees that the outside option (not participating) is always at least as good as the worst outcome from participating.
Pareto Efficiency	Allocations are designed to exhaust all mutually beneficial trades before closing a market. No resource sits idle when a willing buyer and seller could be matched.
Budget Balance	Platform fees and mechanism payouts are designed so the platform's net revenue from running a mechanism is non-negative (weak budget balance) without extracting rents that would deter participation.
Fairness & Non-Discrimination	Matching algorithms do not discriminate on protected characteristics. Scoring functions are transparent, auditable, and explainable to all participants.
Liquidity Design	Mechanisms specifically account for thin markets (few participants) by using reservation prices, automated market makers, and fallback mechanisms to prevent market failure.
Dynamic Consistency	Mechanisms produce consistent outcomes over time — a rule that produces a good outcome today should not produce a bad outcome tomorrow due to strategic adaptation by players.
Transparency	Mechanism rules, scoring weights, and outcome explanations are available to participants. Opaque 'black box' allocations that participants cannot audit are avoided.

2.2 Game Types by Platform Context

Different platform interactions are formalized as distinct game types, each with appropriate mechanism design solutions:

Game Type	Platform Context	Mechanism
Assignment Game	Worker-to-gig matching, resource-to-project matching	Deferred Acceptance (Gale-Shapley) with multi-dimensional scoring

Double Auction	Exchange market — simultaneous buyers and sellers posting prices	Continuous Double Auction (CDA) with price-time priority order book
Vickrey Auction (VCG)	Single-item high-value asset sales, exclusive contracts	Second-price sealed-bid; incentive-compatible, truthful revelation
Combinatorial Auction	Bundled resource or asset packages, multi-resource project staffing	Iterative Combinatorial Auction (ICA) with proxy bidding
Reverse Auction / RFP	Buyer posts requirements; sellers compete for the contract	Score-based sealed-bid reverse auction with price+quality scoring
Negotiated Exchange	Bilateral deal negotiation (Deal Room)	Alternating-offer Rubinstein bargaining model with AI-assisted ZOPA computation
Cooperative Game	Payoff distribution in collectives and cooperatives	Shapley Value computation for fair contribution-based distribution
All-or-Nothing Campaign	Crowdfunding with a minimum threshold	Assurance contract / threshold public goods game
Dynamic Pricing	Platform fee setting, listing boost pricing	Myerson-optimal mechanism with demand curve estimation
Reputation Game	Long-run behavior shaping via reputation scores	Repeated game with discount factor; Folk Theorem equilibrium maintenance

2.3 Zone of Possible Agreement (ZOPA) Engine

For bilateral deal negotiations in the Deal Room, kogi-game computes the Zone of Possible Agreement — the range of prices/terms where both parties' reservation values overlap — using signals from both sides' platform history and market data:

ZOPA Condition	<code>ZOPA ≠ ∅ iff seller_reservation_price ≤ buyer_reservation_price</code>
ZOPA Range	<code>ZOPA = [seller_reservation_price, buyer_reservation_price]</code>

The ZOPA Engine collects: seller's historical deal prices for similar work, buyer's historical spending for similar engagements, current market rate from MatchEngine comparables, and time-pressure signals (deadline proximity, competing offers). Oba presents the ZOPA as a price range guidance panel in the Deal Room, enabling faster convergence to agreement without revealing private reservation prices.

2.4 Shapley Value for Cooperative Distribution

For collectives, cooperatives, and multi-contributor campaigns, kogi-game computes the Shapley Value — the unique fair allocation that assigns each participant their marginal contribution averaged over all possible orderings of the coalition:

Shapley Value

$$\phi_i(\mathbf{v}) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} \times [\mathbf{v}(S \cup \{i\}) - \mathbf{v}(S)]$$

In practice, exact Shapley computation is NP-hard for large coalitions, so kogi-game uses stratified Monte Carlo sampling with $O(n \log n)$ convergence for cooperative entities up to 50 members, and approximation algorithms (Group Shapley) for larger federations. The result drives patronage distribution in cooperatives and contribution-weighted payouts in campaigns.

3. ALLOCATION ENGINE (AllocEngine)

3.1 Purpose

The AllocEngine is the computational core that takes the output of the MatchEngine (ranked candidate pairs) and the rules specified by a given mechanism, and produces a binding allocation: who gets what resource/opportunity, at what price, under what terms, and in what order. It is the 'solver' that turns preferences and bids into outcomes.

3.2 Resource Allocation

3.2.1 Resource Types

Resource Category	Platform Instances	Allocation Mechanism
Human Labor	Worker time, contractor availability, gig capacity	Assignment market — Deferred Acceptance with skill + reputation scoring
Financial Capital	Funds from investors, donors, lenders, campaigners	Budget-constrained combinatorial allocation with portfolio fit scoring
Compute Resources	Platform compute credits, GPU time, API quotas	First-price auction + priority queue for real-time allocation
Intellectual Assets	Templates, playbooks, designs, code modules, datasets	Posted-price marketplace; recommendation-driven discovery
Physical Resources	Equipment, materials, real estate (future)	Lease/rental auction with availability scheduling
Platform Attention	Feed placement, recommendation slots, promoted listings	Ad-auction style with quality-adjusted bid ($QScore \times CPM$)
Cooperative Commons	Shared tools, pooled licenses, mutual aid funds	Equal-access commons with governance-controlled rationing

3.2.2 Labor Allocation — Deferred Acceptance Protocol

The primary mechanism for worker-to-opportunity allocation is a Deferred Acceptance (DA) algorithm — a stable matching mechanism proven to produce stable outcomes where no worker-opportunity pair would both prefer to be matched to each other over their current assignment:

1. Opportunities (gigs, contracts, RFPs) post their requirements and scoring weights
2. Workers apply to opportunities, optionally in rank-ordered preference lists
3. AllocEngine runs DA: opportunities provisionally accept their highest-scored applicants, reject others
4. Rejected workers apply to their next-preferred opportunity
5. Process repeats until no rejected applicant remains — stable matching reached

6. Final assignments reported to kogi-marketplace; escrow instructions generated for kogi-bank

3.2.3 Capital Allocation — Portfolio Budget Engine

When an investor, donor, or funder commits capital, the AllocEngine runs a portfolio budget allocation to determine optimal deployment across competing opportunities:

Budget Allocation

$$\max \sum_i (\text{expected_return}_i \times \text{allocation}_i) \quad \text{s.t.} \quad \sum_i \text{allocation}_i \leq \text{budget}, \quad \text{allocation}_i \geq \text{min_ticket}_i$$

Inputs: investor preference vector (sector, stage, instrument type, risk tolerance), portfolio fit scores from MatchEngine, expected return signals from AnalyticsEngine (cashFlow, productivity, portfolio health), and diversification constraints from the investor's risk policy. Output: ranked investment allocation plan presented to the investor as an Oba-generated recommendation card with one-click execution.

3.3 Mechanism-Specific Allocation Rules

3.3.1 Bid Allocation

Bid Type	Allocation Rule
Open Ascending Bid (English Auction)	Highest bidder at close wins; price = their bid. Used for asset sales and exclusive contract awards.
Sealed First-Price Bid	Highest bidder wins; pays their bid. Used in sealed RFP competitions and blind auctions.
Sealed Second-Price (Vickrey)	Highest bidder wins; pays second-highest bid. Incentive-compatible; dominant strategy is truthful. Used for high-value, single-item allocations.
Combinatorial Bid	Bidder bids on bundles of items. Winner determined by revenue-maximizing allocation over all bids. Used for multi-resource project staffing packages.
Reverse Bid (Buyer Auction)	Lowest qualifying bid wins the buyer's contract. Price = winning bid. Qualification assessed by quality score threshold.
Proxy Bid	Bidder sets max price; system bids incrementally on their behalf up to max. Maintains bidder privacy while pursuing optimal outcome.

3.3.2 Offer Allocation

Offer Type	Allocation Rule
Posted Price (Take-or-Leave)	Single fixed price. First buyer to accept gets the item. No negotiation. Used for standard marketplace listings.

Declining Price (Dutch Auction)	Price starts high and decreases over time. First buyer to accept at current price wins. Creates urgency for buyers.
Bundle Offer	Multiple items offered as a package at a discount. All-or-nothing acceptance. Used for asset portfolios, template bundles.
Subscription Offer	Ongoing access to a portfolio component or service stream at a recurring price. First-come first-served up to capacity limit.

3.3.3 Deal Allocation

Deal Type	Allocation Rule
Bilateral Negotiated Deal	ZOPA Engine guides price discovery; offer/counter-offer protocol. Deal closes when both parties accept simultaneously or one party accepts the last standing offer.
Tripartite Deal	Three-party deal (e.g., worker + client + platform). Each pair has bilateral preferences; AllocEngine finds a jointly Pareto-improving allocation.
Multi-Party Consortium Deal	N-party collective purchase or investment. Each party has a budget share; AllocEngine maximizes collective utility subject to each party's share constraint.
Conditional Deal	Deal terms contingent on a future event (milestone, funding close, regulatory approval). AllocEngine tracks the condition and executes allocation on trigger.

3.3.4 Request / RFP Allocation

Request Type	Allocation Rule
Open RFP	Request broadcast to all qualified workers/vendors. AllocEngine scores all proposals against buyer's scoring rubric. Buyer selects from ranked shortlist.
Restricted RFP	Request sent only to a pre-qualified vendor pool (curated by MatchEngine or buyer). Same scoring; restricted supply side.
Budget-Capped RFP	All proposals at or below budget ceiling scored; cheapest acceptable proposal wins (reverse auction). Scope quality must meet minimum threshold.
Blind RFP	Buyer identities and budgets are anonymized to prevent strategic pricing. Only skills, timeline, and scope are revealed. Promotes honest proposal pricing.

3.3.5 Contract, Gig & Task Allocation

Entity	Allocation Rule
--------	-----------------

Contract	Bilateral assignment after RFP or direct hire. AllocEngine generates milestone payment schedule, assigns escrow plan, and links to PortfolioSystem project as a ResourceAllocation.
Gig	Direct fixed-price assignment, typically from a Proposal or Offer. Fast-track: MatchEngine proposes → buyer accepts → escrow funded in one step.
Task	Sub-unit of a Contract or Gig. Allocated to a worker within an already-agreed engagement scope. No separate bidding; assignment governed by contract terms.
Bounty Task	Open task with a fixed reward posted publicly. First worker to meet acceptance criteria wins the bounty. AllocEngine validates delivery before releasing funds.

3.4 Platform Attention Allocation

Feed placement, recommendation slot prominence, and marketplace discovery ranking are themselves allocation problems. kogi-game governs these 'attention markets' using quality-adjusted score auctions:

Quality Score (QScore)	$\text{QScore} = \text{RelevanceScore} \times 0.40 + \text{ReputationScore} \times 0.30 + \text{EngagementScore} \times 0.20 + \text{FreshnessScore} \times 0.10$
Rank Score (for promoted listings)	$\text{RankScore} = \text{QScore} \times \text{bid_per_impression}$

Organic rankings use QScore alone. Promoted slots use RankScore, ensuring that poorly-quality listings cannot simply outbid high-quality ones to dominate placement. This protects ecosystem health by requiring minimum quality thresholds for promotion participation.

4. MATCHING ENGINE

4.1 Overview

The MatchEngine (kogi-engine sub-system 8) performs multi-dimensional scoring to match any MatchSubject — a worker, portfolio component, resource, investor, or analytics artifact — against any other MatchSubject or pool of subjects. kogi-game extends the base MatchEngine with game-theoretic scoring layers, market-aware boosts, and mechanism-specific match operations for each transaction type in the platform.

4.2 Match Subject Taxonomy

MatchSubject Type	Sub-types	Key Discriminants
UserSubject	Worker, Investor, Donor, Contributor, Collaborator, Buyer, Hirer	role, persona, skills[], interests[], budget, location, reputation_score, availability
ComponentSubject	Portfolio, Program, Project, Resource, Asset, Artifact	kind, status, requiredSkills[], budget, tags[], health_score, visibility
ResourceSubject	Compute, License, Equipment, Template, Dataset	resourceType, capacity, availability, cost, location, tags[]
TransactionSubject	Bid, Offer, Deal, Request, Proposal, Contract, Gig, Task	transaction_type, price, scope, deadline, required_skills[], budget, status
FinancialSubject	Equity round, Campaign, Bond, Revenue-share note	instrument_type, raise_amount, stage, sector, expected_return, risk_level
AnalyticsArtifact	Recommendation, Search result, Index entry, Filter result	artifactType, affinity_tags[], score, freshness

4.3 Core Scoring Dimensions & Weights

Dimension	Default Weight	Computation	Overridable?
Tag Similarity	40%	Jaccard similarity of subject tag sets	Yes — per mechanism type
Attribute Affinity	30%	Weighted overlap of shared attribute key-value pairs	Yes — per mechanism type
Persona Alignment	20%	Persona-to-component-type affinity matrix lookup	Partial — base matrix fixed

Signal Strength	10%	Analytics signal from interaction history and engagement events	No — engine-determined
-----------------	-----	---	------------------------

4.4 Mechanism-Specific Match Operations

4.4.1 Worker-to-Opportunity Matching

Triggered by: new gig/contract posting, RFP broadcast, or worker's Oba morning briefing.

Scoring Dimension	Weight	Data Source
Skills Coverage	35%	Worker skill tags vs. opportunity requiredSkills — exact + fuzzy semantic match
Reputation Score	25%	Worker's historical ratings, completion rate, dispute rate
Budget Compatibility	20%	Worker's rate range vs. opportunity budget — no match if rate > 2× budget ceiling
Availability Alignment	10%	Worker's calendar availability vs. opportunity start/end dates
Location / Timezone Compatibility	5%	Timezone overlap for synchronous collaboration requirements
Past Collaboration Boost	5%	Bonus if worker has worked with this buyer before and both rated positively

4.4.2 Investor-to-Campaign / Round Matching

Triggered by: new campaign/equity round posted, or investor's campaign browsing session.

Scoring Dimension	Weight	Data Source
Portfolio Fit (Sector / Stage)	35%	Investor's portfolio preferences vs. campaign's sector and stage tags
Financial Instrument Preference	25%	Investor's instrument history (equity vs. revenue-share vs. bond) vs. campaign type
Risk Profile Alignment	20%	Investor's risk tolerance vs. campaign's PortfolioHealthScore and RiskEngine output
Return Signal	10%	Expected return estimate from AnalyticsEngine (cashFlow delta, comparable outcomes)

Social Proof	10%	Number of existing investors the target investor follows who have already backed the campaign
--------------	-----	---

4.4.3 Resource-to-Project Matching

Triggered by: project's resource gap detection by RiskEngine (OptimizationPlan), or manual resource search.

Scoring Dimension	Weight	Data Source
Functional Fit	40%	Resource capability tags vs. project requiredSkills / requiredResources
Cost Within Budget	30%	Resource cost vs. project ResourceAllocation budget remaining
Availability	20%	Resource availability window vs. project timeline (GraphEngine CPM)
Quality Signal	10%	Resource's historical usage ratings and delivery success rate

4.4.4 Collaborator Discovery Matching

Triggered by: user searching for co-founders, contributors, co-creators, or community collaborators.

Scoring Dimension	Weight	Data Source
Complementary Skills	30%	Jaccard distance of skill sets — seeks differentiation, not duplication
Shared Interests / Goals	30%	Interest tag overlap weighted by interaction depth
Community Graph Proximity	25%	Second-degree social graph distance via GraphEngine relationship edges
Portfolio Compatibility	15%	Component kind and status alignment — both in compatible lifecycle stages

4.4.5 Portfolio Asset / Artifact Trading Matching

Triggered by: user browsing marketplace, or asset listed for sale.

Scoring Dimension	Weight	Data Source
Category / Tag Relevance	40%	User's dominant categories and interest tags vs. asset tags

Price Affinity	30%	Buyer's typical spend range vs. asset price — strong signal from purchase history
Quality Score	20%	Asset's PortfolioHealthScore, ratings, download/usage count
Social Proof	10%	Number of people in buyer's network who own or rated this asset

4.5 Persona-Aware Match Boosting

Persona	Boost Applied	Rationale
PowerUser	Asset and Program component matches $\times 1.2$	Power users extract more value from complex components
Investor (role)	Equity and revenue-share campaign matches $\times 1.3$	Primary motivation is financial return
Explorer	Diversity bonus — surfaces less-obvious but high-potential matches	Explorers seek discovery, not confirmation
Specialist	Domain-exact tag matches $\times 1.25$	Specialists want depth, not breadth
Collaborator	Network-adjacent matches $\times 1.2$ — people known by people they know	Collaborators build through trust networks
Newcomer	Trending and well-rated items prioritized — cold-start protection	Insufficient history to personalize; use global signals
ValueSeeker	Price-efficiency score boosted $\times 1.2$	Maximizing return per dollar is primary driver

4.6 Match Confidence & Explainability

Every match produced by the MatchEngine carries a confidence score (0.0–1.0) computed from the completeness of input signals and the margin between the top match and the second-best match. Matches with confidence < 0.4 are flagged for human review or Oba clarification. Every match result also carries an explanation vector — the weighted contribution of each scoring dimension — enabling the UI to display 'Why we matched you: Skills 62%, Budget 22%, Availability 16%' to both parties.

5. RECOMMENDATION ENGINE

5.1 Architecture Overview

The RecommendationEngine is the platform's content and opportunity discovery system. Where the MatchEngine finds the best fit for a specific defined need, the RecommendationEngine proactively surfaces things users did not know to search for — portfolio components to follow, marketplace listings to buy, collaborators to connect with, investments to consider, campaigns to back — based on behavioral signals, interaction history, and social graph proximity.

5.2 User Profile & Behavioral Signals

Profile Field	Description
interaction_history	Chronological log of all platform interactions: views, clicks, purchases, ratings, bookmarks, dismisses
preference_vector	Normalized weighted sum over all interaction categories — continuously updated after each interaction
dominant_categories	Top-3 categories by cumulative interaction weight — drives content ordering
session_context	Current device, location, time-of-day, referrer — contextual adaptation
persona	Derived behavioral archetype: PowerUser, CasualBrowser, Explorer, EarlyAdopter, ValueSeeker, Specialist, Collaborator, Newcomer
explicit_preferences	User-declared interests, preferred content types, blocked topics, notification preferences
social_graph	Follow, subscribe, collaborate, and team relationships — second-degree social signals
financial_preferences	Investment preferences, budget ranges, risk tolerance — for financial instrument recommendations
portfolio_context	Active portfolio components and their tags — context for 'things relevant to my current work'

5.3 Recommendation Methods

5.3.1 Collaborative Filtering

Finds users with similar interaction histories using cosine similarity on preference vectors. Items interacted with by similar users that the target user has not yet seen are surfaced, weighted by the collaborating user's similarity score and interaction weight. Best for: discovering popular items in the user's interest space that they have not found independently.

Collaborative Score

$$cf_score(u,i) = \frac{\sum_{v \in neighbors(u)} sim(u,v) \times interaction_weight(v,i)}{|neighbors(u)|}$$

5.3.2 Content-Based Filtering

Compares the user's preference vector against item feature vectors derived from tags, category, and attributes. Items with highest cosine similarity to the user's accumulated profile are returned with a recency boost for recently created items. Best for: recommending items highly similar to what the user already engages with.

Content Score

$$cb_score(u,i) = cosine(preference_vector(u), feature_vector(i)) \times recency_boost(i)$$

5.3.3 Hybrid Scoring (Default)

Blends collaborative (40%), content-based (40%), and global popularity (20%) scores. Contextual boosting applied on top based on current session context (device, time-of-day, location). Cold-start fallback to trending+popular when interaction history is insufficient.

Hybrid Score

$$hybrid(u,i) = 0.40 \times cf_score + 0.40 \times cb_score + 0.20 \times popularity_score + context_boost$$

5.4 Recommendation Surfaces

Surface	Recommendation Type	Trigger
Home Dashboard Feed	Mixed: portfolio updates, new opportunities, community content, trending assets	Page load; real-time push on new matching events
Marketplace Discovery	Listings, workers, assets, and campaigns matching buyer's profile	Browse session; after purchase event (complementary goods)
Exchange Discovery	Financial instruments, equity rounds, and trading opportunities	Exchange session; after investment event
Collaborator Suggestions	Users with complementary skills or shared interests	Project creation; sprint blocker detected by RiskEngine
Resource Suggestions	Tools, templates, datasets, and services for active projects	Project creation; resource gap detected in OptimizationPlan
Morning Briefing Cards	Personalized action items: opportunities, follow-ups, risk alerts	Oba morning briefing generation; daily digest
Invest/Back Suggestions	Campaigns and rounds matching investor's portfolio preferences	Campaign creation; investor browsing session

Portfolio Expansion	New portfolio component types and templates to start	After project completion; portfolio health review
Community Space Suggestions	Spaces and communities matching user's tags and persona	Community browse; new member onboarding
Explore Feed	Directional expansion: 'based on X, explore Y and related areas'	Explicitly triggered by user; auto-surfaced for Explorer persona

5.5 Feedback Loop & Continuous Improvement

The RecommendationEngine maintains a continuous feedback loop. Every user interaction with a recommendation — click, view, dwell, purchase, bookmark, dismiss — is recorded as a UserInteraction and fed back into the profile, adjusting the preference vector and updating collaborative similarity scores. The feedback loop has the following latency profile:

Feedback Event	Update Latency	Effect
Purchase / Rate	< 2s	Immediate: preference_vector += 5.0 × item feature vector; nearest-neighbor graph updated
Bookmark / Share	< 2s	Immediate: preference_vector += 3.0 × item feature vector
Click / View	< 10s (batched)	Near-real-time: preference_vector += 1.5 × item feature vector
Dismiss	< 10s (batched)	Near-real-time: preference_vector -= 2.0 × item feature vector; item suppressed for 30 days
Dwell (>30s)	< 30s	Background: preference_vector += 1.0 × item feature vector
Session-level re-ranking	Every 5 minutes	Background: re-score all cached recommendation lists with updated profile
Full profile rebuild	Daily	Batch: recompute all similarity scores, dominant categories, persona re-inference

6. PERSONALIZATION ENGINE

6.1 Purpose

The PersonalizationEngine extends the RecommendationEngine with platform-level personalization: how content is displayed, how users are segmented into cohorts, how A/B experiments are assigned and tracked, how the multi-armed bandit selects between content variants, and how the persona lifecycle evolves over time. Where the RecommendationEngine answers 'what should this user see?', the PersonalizationEngine answers 'how should they see it?'

6.2 Persona System

Persona	Behavioral Profile	UI Config	Score Weights
PowerUser	High frequency, advanced features, broad exploration	Compact density, RecencyFirst, 50 feed items, analytics sidebar, advanced filters	Risk $\times$ 1.10 (tighter guardrails)
CasualBrowser	Low engagement, short sessions, trending content	Spacious, TrendingFirst, 20 feed items, auto-expand items, suppress low-engagement	Risk $\times$ 0.85, Productivity $\times$ 0.90 (reduced noise)
Explorer	Diverse topics, high discover/explore engagement	Standard, RelevanceFirst, 40 feed items, diversity bonus, highlight new content	Collaboration $\times$ 1.15 (compensates lower natural score)
EarlyAdopter	Rapid new item adoption, beta features	Compact, RecencyFirst, highlight new, novelty score boost	No persona-specific weight adjustment
ValueSeeker	Price/ROI sensitive, deal-oriented	Standard, ValueFirst, price-efficiency score boosted	CashFlow $\times$ 0.90 (amplifies cashflow sensitivity)
Specialist	Deep domain focus, low breadth / high depth	Standard, RelevanceFirst, domain-specific recs, 10K row query cap, analytic hints	No persona-specific weight adjustment
Collaborator	High social engagement, groups, sharing	Standard, PersonalFirst, collaboration signal boost	No persona-specific weight adjustment
Newcomer	Cold-start, low history	Spacious, TrendingFirst, 20 feed items, cold-start recs	All signals $\times$ 0.80 (low confidence, prefer global signals)

6.3 User Preferences System

Users can explicitly override any persona-derived default through the Preferences system — a structured set of user-declared configuration options that always take precedence over inferred defaults:

Preference Category	Options	Effect on Engine
Content Ordering	RecencyFirst, RelevanceFirst, TrendingFirst, ValueFirst, PersonalFirst	Overrides persona's default ordering in all feed surfaces
Feed Density	Compact, Standard, Spacious	Overrides persona's layout; affects all card and list views
Interest Topics	Explicit include and exclude topic tags	Boosts included tags $\times 1.5$; filters out excluded tags entirely
Notification Preferences	Which event types trigger push/email; frequency (real-time, daily, weekly digest)	Governs Oba briefing card categories and alert thresholds
Financial Preferences	Preferred instrument types, min/max deal sizes, sector interests, risk appetite (1–5)	Weights FinancialSubject matches and IncentiveEngine financial nudges
Collaboration Preferences	Open to collaborator suggestions (Y/N), preferred team size, remote/local preference	Governs Collaborator Discovery surface and community space suggestions
Privacy Preferences	Profile visibility, portfolio visibility, anonymization on analytics exports	Governs what signals are observable by the RecommendationEngine and shared with third parties

6.4 A/B Experiment Framework

The PersonalizationEngine manages platform-wide A/B experiments for mechanism and UI variants:

- Experiment definition — id, name, variants (each with traffic allocation weight), status (draft / active / paused / completed), hypothesis, success metric
- Deterministic assignment — hash-based user-to-variant assignment ensures every user always gets the same variant for a given experiment; prevents experience fragmentation
- Exposure tracking — records which users were exposed to which variants, with timestamp and session context
- Conversion recording — records when an exposed user completes the experiment's target action (click, purchase, deal close, etc.)
- Statistical analysis — Bayesian A/B test with credible intervals; lift calculation per variant; minimum detectable effect sizing
- Guard rails — automatic experiment pause if any variant produces anomalous negative signals (error rate, churn rate, dispute rate above threshold)

6.5 Multi-Armed Bandit (UCB1)

For content slot optimization and mechanism variant selection — where the best option is unknown and must be learned through exploration — the PersonalizationEngine implements the UCB1 (Upper Confidence Bound) algorithm:

UCB1 Score

$$\text{UCB1}(\text{arm}) = \text{avg_reward}(\text{arm}) + \sqrt{(2 \times \ln(\text{total_pulls}) / \text{arm_pulls})}$$

Arms with zero pulls are always selected first (pure exploration priority). After sufficient pulls, exploitation naturally dominates. This balances exploration of untested variants against exploitation of known-good variants, converging to the optimal variant without requiring pre-specified exploration schedules.

6.6 Persona Lifecycle Management

Personas evolve dynamically as user behavior changes — they are not static labels:

7. Build — initial persona inference from interaction history (minimum 20 interactions) and available profile data
8. Confirm — persona stabilizes after N consecutive sessions without drift crossing the re-assignment threshold
9. Drift Detection — current interaction pattern compared against persona archetype thresholds; drift score computed
10. Update — new persona label assigned if drift score exceeds 0.35 threshold; full history log maintained
11. Hysteresis — persona downgrade requires stronger evidence than upgrade to prevent oscillation
12. Manual Override — user can pin a persona label in preferences, preventing automatic re-assignment

7. INCENTIVE ENGINE (IncentiveEngine)

7.1 Purpose & Design

The IncentiveEngine governs the long-run behavior of all platform participants by designing, tracking, and administering a multi-dimensional incentive system that rewards value-creating behaviors, penalizes value-destroying behaviors, and maintains equilibrium conditions necessary for a healthy multi-sided market. It is the platform's mechanism for shaping participant behavior across repeated interactions over time.

The IncentiveEngine draws on Repeated Game Theory — specifically the Folk Theorem, which establishes that in infinitely repeated games with patient players, cooperative equilibria (where all participants behave well) can be sustained through credible threats of punishment for defection. The platform acts as the long-run game coordinator, maintaining the threat of reputation penalties and access restrictions that make good behavior the dominant strategy even for self-interested actors.

7.2 Reputation System

7.2.1 Reputation Score Architecture

Every user and organizational entity has a multi-dimensional ReputationScore — not a single number, but a vector of scores across the interaction domains relevant to them:

Dimension	Measures	Decay / Update Rule
Delivery Reliability	Contract / gig completion rate; on-time delivery rate; cancellation rate	Rolling 12-month window; recent events weighted 2× older events
Communication Quality	Response time to messages; clarity of scope documentation; dispute rate	Per-interaction rating; Bayesian update with prior = platform average
Work Quality	Client ratings 1–5 across Quality, Timeliness, Communication, Value	Weighted average; minimum 5 ratings for public display
Financial Integrity	Invoice accuracy; payment timeliness (for buyers); dispute resolution outcome	Binary flags for fraud, non-payment; decay after 24 clean months
Community Contribution	Post quality ratings; knowledge sharing; mentorship; collaborative deliveries	Accumulated score; decays slowly if inactive (×0.95 per 90 days)
Investment Track Record	Campaign success rate backed; portfolio health of invested components	Rolling score; only applies to investor role
Governance Participation	Vote participation rate in collectives; proposal quality ratings	Cooperative/collective entities only

7.2.2 Composite Reputation Score

Composite Score	$R(u) = 0.30 \times \text{Delivery} + 0.25 \times \text{Quality} + 0.20 \times \text{Communication} + 0.15 \times \text{Community} + 0.10 \times \text{Financial}$
------------------------	--

The composite score produces a single 0–100 number displayed on user and organization profiles. Score bands translate to platform status tiers with associated incentive benefits:

Tier	Score Range	Benefits	Access
Elite	90–100	Featured placement boost $\times 1.5$; reduced platform fees (–20%); early access to beta features; priority dispute resolution	All marketplace, exchange, and campaign types
Trusted	75–89	Featured placement boost $\times 1.2$; reduced platform fees (–10%); verified badge on all listings	All marketplace and exchange types
Established	55–74	Standard placement; standard fees; verified identity badge	All standard marketplace and exchange types
Developing	35–54	Standard placement; standard fees; no promoted listing access until 55+	Standard gigs, proposals, and campaigns only
New	0–34	Cold-start tier; guided onboarding; limited deal sizes; escrow mandatory	Simple gigs and low-value transactions only

7.3 Reward Mechanisms

7.3.1 Platform Credit System (KogiPoints)

KogiPoints (KP) are the platform's internal incentive currency — non-monetary credits earned through value-creating behaviors and redeemable against platform fees, promoted listing boosts, and premium features:

Earning Event	KP Awarded	Rationale
Complete a gig / contract (seller)	50–500 KP (proportional to contract value)	Core labor market activity
Complete a deal as buyer (first hire of a worker)	100 KP	Buyer activation incentive
Receive 5-star rating	25 KP	Quality signaling reward

Publish a portfolio component to marketplace	75 KP	Supply-side contribution
Back a campaign (first backing)	50 KP	Campaign liquidity incentive
Successful referral (new user completes first deal)	250 KP	Network growth reward
Complete onboarding profile (skills + portfolio + availability)	100 KP one-time	Information quality incentive
Write a detailed review (>100 words)	15 KP	Review quality incentive
Participate in cooperative governance vote	10 KP	Governance participation
Mentor a Newcomer (guided interaction)	50 KP	Community contribution
Resolve a dispute without escalation	30 KP	Platform health reward

Redemption	KP Cost	Effect
Platform fee discount (10%)	200 KP per invoice	Direct financial benefit
Promoted listing boost (7 days)	500 KP	Increased discovery placement
Priority matching queue (30 days)	1000 KP	Top-of-list for incoming RFP matches
Campaign boost (7 days)	750 KP	Campaign featured in discovery feed
Exchange order fee waiver (1 trade)	150 KP	Per-transaction fee reduction

7.3.2 Streak & Consistency Incentives

Streaks reward sustained behavioral consistency — a form of dynamic incentive that increases marginal reward for continued participation:

- Activity streak — earn bonus KP multiplier for each consecutive week with at least one completed transaction ($\times 1.05$ per week, cap $\times 2.0$)
- Response rate streak — maintain $>90\%$ response rate for 30+ consecutive days: promoted in worker discovery ranking
- Review streak — write a quality review after every engagement for 5+ consecutive engagements: earn Reviewer badge

- Campaign streak — successfully complete 3+ sequential campaigns as creator: earn Campaign Creator badge with fee discount
- Governance streak — participate in every cooperative vote for 6+ consecutive months: earn Governance Champion badge

7.3.3 Subscription & Loyalty Incentives

Portfolio component creators can offer subscriber-specific incentives, governed by the IncentiveEngine:

- Subscriber discounts — creators set percentage discounts for subscribers on new paid components (the user 1 / user 2 real-estate playbook model)
- Early access tiers — subscribers get 24-hour early access before public listing
- Loyalty pricing — subscribers who have maintained subscription for 12+ months receive a locked-in price regardless of future price increases
- Affiliate commissions — creators can define affiliate link programs; referred buyers tracked by ProviderSystem; commissions automatically paid from kogi-bank

7.4 Penalty Mechanisms

The IncentiveEngine enforces penalties for value-destroying behaviors to maintain the repeated-game equilibrium:

Behavior	Penalty	Duration
Contract cancellation (seller-initiated without cause)	–25 Delivery Reliability score; –50 KP; listing visibility reduced for 30 days	30-day visibility penalty; score impact decays over 12 months
Non-payment (buyer-initiated without dispute)	–25 Financial Integrity score; account flagged; escrow mandatory for 6 months	Account flag for 6 months; cleared on 3 clean payments
Fraudulent listing (counterfeit asset, misleading scope)	Listing removed; –50 overall reputation score; 30-day suspension	Permanent listing removal; appeal process available
Dispute filed and lost (either party)	–10 overall reputation score	Score impact decays over 12 months
Repeated late delivery (3+ in 6 months)	–15 Delivery Reliability; Developed tier downgrade	Tier downgrade for 90 days; restored on improved delivery
Platform abuse (spam, fake reviews, bot activity)	Account suspension; KP forfeiture; IP-level block after 3 offenses	1st: warning; 2nd: 30-day suspension; 3rd: permanent

7.5 Market Liquidity Incentives

Thin markets — categories with few buyers and sellers — can cause market failure. The IncentiveEngine detects thin market conditions (< 10 active listings in a category, < 5 active buyers in a period) and activates liquidity incentives:

- Supply-side bonus — new listings in thin categories earn 2× KP for the first 30 days
- First-mover bonus — the first worker to complete a gig in a new category earns a Pioneer badge and 500 KP
- Buyer activation — platform temporarily reduces or waives listing fees in thin categories to increase supply
- Cross-category recommendations — MatchEngine boosts adjacent category workers into thin category searches with a skill-transfer score
- Campaign seeding — platform can seed small initial contributions to new campaigns in thin funding categories to trigger social proof effects

8. ANALYTICS & INTELLIGENCE LAYER

8.1 PortfolioSignal — The Universal Health Metric

The PortfolioSignal is the four-dimensional score that every platform entity — user, portfolio component, organization, marketplace listing — continuously emits. It is the primary input to the RiskEngine, AnalyticsEngine, MatchEngine, and IncentiveEngine:

Dimension	Range	Sources	Game Role
Productivity	0–100	Task completion velocity, throughput, sprint burn rate, gig completion rate	Supply quality signal; drives MatchEngine labor scoring
CashFlow	0–100	Invoice payment rate, incoming transaction velocity, budget utilization, escrow release rate	Financial health signal; drives IncentiveEngine financial nudges
Collaboration	0–100	Response rate, co-contributor density, community engagement, dispute rate	Network quality signal; drives Collaborator matching boost
Risk	0–100 (lower is better)	Error rates, overdue items, blocker count, anomaly flags	Risk signal; drives RiskEngine status (green/amber/red)

8.2 PortfolioHealthScore

Health Score	$H = 0.30 \times \text{Productivity} + 0.30 \times \text{CashFlow} + 0.20 \times \text{Collaboration} + 0.20 \times (100 - \text{Risk})$
--------------	--

Health Status	Score Range	Engine Actions
Green	≥ 80	Standard recommendations; no intervention required
Amber	60–79	Oba generates OptimizationPlan; specific improvement suggestions surfaced in morning briefing
Red	< 60	High-priority alert; Oba triggers Stabilization Playbook; opportunity matching temporarily narrowed to lower-risk engagements

8.3 RiskEngine Outputs for Game Mechanics

The RiskEngine produces RiskOptimizationPlans that directly feed into allocation and matching decisions:

Risk Condition	Threshold	Allocation / Matching Response
High risk (risk ≥ 70)	risk signal	MatchEngine applies risk discount on new high-value matches; IncentiveEngine reduces promoted listing eligibility
Low cashflow (cashFlow < 50)	cashFlow signal	Oba prioritizes invoice recovery and deal completion in briefing; RFP match score temporarily downweighted (buyer may not pay)
Low productivity (productivity < 50)	productivity signal	New gig/contract matching paused until sprint workload rebalanced; Oba flags WIP overload
Low collaboration (collaboration < 50)	collaboration signal	Collaborator suggestions increased in feed; community space recommendations amplified
Health status Red	composite	New campaign backing temporarily restricted; escrow terms tightened; reputation tier review triggered

8.4 GraphEngine Integration

The GraphEngine powers the dependency and impact analysis that makes kogi-game's allocation decisions portfolio-aware — not just point-in-time transactions, but decisions informed by the full graph of what depends on what:

- Critical Path Analysis (CPM) — identifies the longest delivery path through a project's dependency graph; AllocEngine uses critical path nodes to prioritize resource allocation to unblock the schedule
- Impact Report — before approving a new contract or resource allocation, GraphEngine computes the downstream impact on all dependent portfolio components
- Cycle Detection — prevents circular dependencies in multi-party deals and sub-contractor chains that could create allocation deadlocks
- Topological Scheduling — AllocEngine uses topological sort to sequence milestone payments in multi-milestone contracts so that payments respect delivery dependencies
- Change Propagation — when a portfolio component's status changes (delayed, cancelled, completed), GraphEngine propagates the impact to all dependent components and triggers re-matching for affected resource gaps

8.5 Streaming Analytics Pipeline

All game events are streamed through the kogi-engine analytics pipeline in real-time:

Kafka Topic	Events	Consumer / Purpose
game.matches	match.computed, match.accepted, match.rejected, match.expired	AnalyticsEngine: match quality tracking; RecommendationEngine: feedback loop update
game.allocations	alloc.proposed, alloc.accepted, alloc.failed, alloc.cancelled	AllocEngine: market clearing analysis; IncentiveEngine: completion/cancellation scoring
game.bids	bid.placed, bid.accepted, bid.rejected, bid.expired, bid.withdrawn	AllocEngine: auction clearing; PriceEngine: market rate estimation
game.deals	deal.initiated, deal.agreed, deal.milestone_completed, deal.disputed, deal.closed	IncentiveEngine: reputation updates; kogi-bank: escrow instructions
game.reputation	score.updated, tier.changed, penalty.applied, reward.credited	UI feed: reputation notifications; MatchEngine: score cache invalidation
game.campaigns	campaign.backed, campaign.funded, campaign.failed	IncentiveEngine: backer KP; AllocEngine: campaign allocation execution
game.incentives	kp.earned, kp.redeemed, streak.updated, badge.awarded	UI feed: achievement notifications; Oba: briefing cards

9. TRANSACTION-SPECIFIC GAME DESIGNS

9.1 Bid / Offer Game

Bids and Offers are the atomic price-discovery instruments of the Kogi Exchange. kogi-game governs the bid/offer lifecycle with mechanism rules that ensure price efficiency and prevent gaming:

Mechanism	Applies To	Rules
Open Ascending (English)	Public asset auctions, featured contract awards	Minimum bid increment: 2% of current price or \$1 (whichever greater). Auction extension: if bid placed in last 5 minutes, extend by 5 minutes (anti-sniping). Reserve price: seller may set a hidden reserve.
Second-Price Sealed (Vickrey)	High-value exclusive contracts, single premium asset sales	All bids submitted simultaneously. Winner = highest bidder; price = second-highest bid. Dominant strategy: bid truthfully.
Continuous Double Auction	Exchange market — labor, commodities, financial instruments	Order book matching: price-time priority. Market order: execute at best available price. Limit order: execute only at specified price or better.
Reverse Auction	Buyer-initiated RFP competition	Seller bids are scope+price packages. Buyer's scoring rubric: price weight + quality weight (configurable). Minimum quality threshold must be met; lowest weighted score wins.
Dutch Auction	Time-sensitive asset offers, expiring resource listings	Starting price set by seller. Price decrements every N hours by a defined step. First buyer to accept wins.

9.2 Deal Room Game

The Deal Room is the platform's bilateral negotiation environment. kogi-game provides the Deal Room with a structured game protocol that guides both parties toward agreement without requiring external arbitration:

13. Game opens: Seller posts Proposal (scope + price + timeline). Buyer opens Deal Room.
14. ZOPA Engine computes price range guidance from historical market data (displayed privately to each party as 'market range').
15. Alternating-offer protocol: Buyer can Accept, Counter, or Request Modification. Each counter-offer resets a response timer.

16. Oba facilitates: drafts structured counter-offers for both parties, highlights ZOPA proximity, and suggests standard deal terms from platform templates.
17. Commitment devices: either party can issue a 'final offer' flag — accepted or negotiation restarts. Prevents endless iteration.
18. Escrow provision: once terms agreed, AllocEngine generates the escrow plan; kogi-bank provisions the escrow account.
19. Game closes: deal signed, escrow funded, project kicks off in kogi-portfolio.

9.3 Proposal Scoring

Proposals submitted to an RFP or open marketplace are scored by the AllocEngine using a configurable scoring rubric. The default rubric weights:

Scoring Dimension	Default Weight	Buyer Can Adjust?	Signal Source
Scope Coverage	25%	Yes ($\pm 10\%$)	AI-assessed: proposal text coverage of requirement tags
Price Competitiveness	30%	Yes ($\pm 15\%$)	Ratio of bid price to median comparable market rate
Reputation Score	25%	Yes ($\pm 10\%$)	MatchEngine: worker's composite ReputationScore
Timeline Commitment	10%	Yes ($\pm 5\%$)	Milestone schedule vs. buyer's deadline constraints
Portfolio Evidence	10%	Yes ($\pm 5\%$)	Relevance of worker's portfolio components to scope

Buyers who consistently apply extreme weightings (e.g., 90% price / 10% quality) are flagged by the IncentiveEngine — statistically, extreme price-weighting correlates with higher dispute rates. Oba suggests rebalancing the rubric toward better outcomes.

9.4 Campaign Game (Threshold Public Goods)

Campaigns are assurance contract games — a form of threshold public goods provision where individual contributions are only realized if the collective threshold is met. kogi-game implements this as follows:

20. Campaign creator sets goal G , deadline D , and type (all-or-nothing or keep-what-you-raise)
21. Backers commit contributions c_i , held in escrow in the Campaign Account in kogi-bank
22. If $\sum c_i \geq G$ before deadline D : campaign succeeds; funds disbursed per milestone schedule

23. If $\sum c_i < G$ at deadline D (for all-or-nothing): all c_i refunded atomically — no backer is worse off
24. Dominant strategy for backers: contribute your true value v_i if $v_i > 0$ and you believe the campaign will succeed (coordination game — social proof and velocity signals reduce coordination failure)
25. IncentiveEngine activates velocity warnings when campaign pace falls behind the required rate: Oba generates targeted outreach to likely contributors

9.5 Cooperative Game (Shapley Distribution)

When a cooperative or collective completes a joint project or closes a financial period, the AllocEngine distributes the surplus using a Shapley Value approximation:

26. Collect contribution_vector for each member: hours contributed, revenue generated, capital contributed, skills deployed
27. Normalize contribution dimensions to a common scale (percentile rank within the cooperative)
28. Compute weighted contribution score per member: $w_1 \times \text{hours} + w_2 \times \text{revenue} + w_3 \times \text{capital} + w_4 \times \text{skills}$
29. Run Monte Carlo Shapley approximation (1,000 samples for ≤ 20 members; Group Shapley for larger groups)
30. Present distribution plan to governance for approval vote
31. On vote passage, AllocEngine generates payroll batch in kogi-bank; executed atomically

9.6 Resource Sharing Game (Commons Management)

Collectives operating shared resource pools (compute credits, licenses, mutual aid funds) face a commons management problem — how to allocate finite shared resources fairly without the Tragedy of the Commons. kogi-game applies a governance-aware commons mechanism:

- Usage quota — each member has a per-period quota determined by their contribution history (contributive justice) and need signals
- Excess transfer — unused quota can be transferred to other members within a period; non-transferable across periods (prevents hoarding)
- Contribution multiplier — members who contribute to the commons pool above their quota earn a multiplier on future quota allocations
- Emergency override — members can request emergency access above quota with a governance fast-track approval (simple majority, 24-hour window)
- Replenishment signal — IncentiveEngine alerts when pool is below 20% capacity; auto-sends replenishment campaign to member group

10. PLATFORM ECONOMICS & FEE DESIGN

10.1 Fee Architecture

Platform fees are themselves a mechanism design problem: fees must be high enough to sustain platform operations and curate quality, but low enough to not deter participation or create incentives to route transactions off-platform. kogi-game designs fees with the following principles: fees scale with transaction value (not fixed), fees are lower for repeat participants (loyalty), and fee revenue is partially recycled into liquidity incentives.

Transaction Type	Fee Structure	Rationale	Loyalty Discount
Gig / Contract (seller fee)	8% of contract value for new sellers; declining to 4% at Elite tier	Covers platform services + escrow; incentivizes quality progression	–1% per tier level (Developing=8%, Established=7%, Trusted=6%, Elite=4%)
Gig / Contract (buyer fee)	2% of contract value (covers payment processing and matching services)	Low buyer fee to maintain demand-side participation	–0.5% at Trusted+; waived at Elite
Marketplace listing sale	6% of sale price	Platform facilitation + discovery services	–1.5% at Elite tier
Exchange trade	0.5% of trade value (both sides)	Order book maintenance, settlement, anti-fraud	Flat rate; reduced by 50% for KP redemption
Campaign (success fee)	5% of successfully raised funds	Campaign hosting + disbursement services	–1% for cooperative / collective entities
Campaign (equity)	3% + 0.5% carried interest on exit	Reflects higher-value financial facilitation	Fixed; non-discountable (regulatory reasons)
Promoted listing	KP-based or \$5–\$50/week based on category competition	Attention allocation market	KP redemption available
Subscription listing fee	1% of recurring subscription revenue	Ongoing platform services for subscription management	Waived for Elite tier

10.2 Market Rate Engine (PriceEngine)

The PriceEngine maintains a continuously-updated model of market rates for every skill category, resource type, and asset class on the platform. It is used by the ZOPA Engine, MatchEngine, AllocEngine, and Oba to provide pricing guidance to all participants:

Rate Category	Sources	Output
---------------	---------	--------

Skill / Labor Rates	Completed contract prices by skill tag and complexity; worker rate cards; external benchmarks (Upwork, Fiverr integration data)	Median, P25, P75 rates per skill × complexity × region
Asset Prices	Completed marketplace sales by asset type and category; listing price history	Price distribution per asset category; comparable recent sales
Resource Rates	Platform resource pricing; third-party provider pricing (compute, licenses)	Unit cost by resource type
Financial Returns	Completed campaign outcomes, equity exit returns, revenue-share performance	Expected return distribution by instrument type and sector

Market Rate (skill s , complexity c)

`rate(s,c) = median(completed_contracts where skill=s and complexity_band=c, last_90_days)`

10.3 Fee Redistribution

A portion of platform fee revenue is recycled into the incentive economy to sustain liquidity and grow participation:

- Liquidity incentive pool (15% of fee revenue) — allocated as KP bonuses to participants in thin markets and new categories
- Newcomer subsidy pool (5% of fee revenue) — reduced fees and matching bonuses for users completing their first 3 transactions
- Community fund (5% of fee revenue) — administered by a platform governance council; distributed as grants for open community contributions (templates, playbooks, public datasets)
- Cooperative bonus (5% of fee revenue) — additional fee discounts for formally incorporated cooperative entities, recognizing their positive externality contribution to the platform ecosystem

11. OBA AI GAME AGENT

11.1 Oba as Market Advisor

Oba, the Kogi Platform's AI assistant, is the primary interface through which kogi-game's mechanism outputs are delivered to users. Oba does not make allocation decisions — those are made by the AllocEngine and governed by the mechanism rules specified in this document — but it explains, facilitates, and implements those decisions in user-facing natural language. Oba is the game's 'referee announcer': it tells each player what the game state is, what their options are, and what the likely consequences of each option are.

11.2 Game Intelligence Workflows

11.2.1 Match & Opportunity Briefing

Each morning, Oba composes a personalized match briefing for each user, driven by overnight MatchEngine results:

- Top-3 new opportunity matches (gigs, contracts, RFPs) ranked by composite match score — with explanation ('Skills match 85%, Budget compatible, New client with 4.8 reputation')
- Top-3 new asset/resource matches — things to buy, license, or follow based on current project context
- Investor suggestion cards for users with investment preferences and available capital
- Collaborator suggestions for users with active projects in Amber or Red health status

11.2.2 ZOPA Facilitation in Deal Room

When a user enters a Deal Room, Oba activates as negotiation facilitator:

- Computes and presents the ZOPA range: 'Based on comparable deals, market rate for this scope is \$X-\$Y per milestone'
- Drafts structured counter-offers in the user's voice when they specify intent in plain language
- Tracks offer history and highlights trajectory: 'You are \$200 apart — typically deals at this stage close within 2 rounds'
- Suggests standard deal terms (payment schedule, milestone definitions, revision limits) from platform templates
- Flags deal terms that statistically correlate with disputes: 'Scope as written has led to disputes in 23% of similar deals — suggest adding deliverable acceptance criteria'

11.2.3 Proposal Drafting & RFP Response

For sellers responding to RFPs, Oba auto-drafts competitive proposals calibrated to the buyer's scoring rubric:

- Reads the RFP requirements and buyer's scoring weights

- Pulls relevant portfolio components, past work evidence, and skill evidence to populate the proposal
- Computes a Price Competitiveness score: 'Your draft price is at the 45th percentile — competitive but room to price higher given your reputation tier'
- Suggests scope clarifications where the RFP is ambiguous — preventing future disputes

11.2.4 Campaign Intelligence & Velocity Coaching

For campaign creators, Oba runs a continuous velocity monitoring loop:

- Tracks contribution rate vs. required pace to meet goal by deadline
- Predicts success probability: 'At current pace, 67% chance of reaching goal — need to accelerate by \$X/day'
- Identifies high-probability backers from social graph and platform data — drafts personalized outreach
- Suggests campaign boosts (KP redemption for featured placement) when velocity drops below critical threshold

11.2.5 Incentive Status & KP Management

Oba tracks and surfaces each user's incentive status proactively:

- Morning briefing: KP balance, active streaks, next milestone to next reputation tier, pending badge progress
- Post-transaction: 'You earned 125 KP for completing this contract — your Delivery Reliability score increased to 94'
- Tier upgrade notification: 'You have reached Trusted tier — your platform fee has reduced from 7% to 6% effective immediately'
- Streak at-risk warning: 'Your activity streak is at 6 weeks — complete one transaction this week to keep your ×1.30 KP multiplier'

11.2.6 Risk-Adjusted Opportunity Guidance

Oba integrates RiskEngine output into opportunity guidance — steering users away from commitments that would overload already-stressed portfolios:

- If portfolio health is Amber: 'Your current workload is near capacity — this new gig would push your sprint utilization to 130%. Recommend declining or deferring to next sprint.'
- If cashFlow signal is low: 'Two invoices are overdue totaling \$3,400. Recommend recovering these before taking on new unpaid work.'
- If a counterparty has low reputation: 'This buyer's Financial Integrity score is 35 — historically 40% of contracts with this score tier result in late payment. Escrow strongly recommended.'

12. DATA ARCHITECTURE

12.1 Core Tables

Table	Description	Key Columns
reputation_scores	Per-user/org multi-dimensional reputation records	id, entity_id, entity_type, delivery, quality, communication, community, financial, composite, tier, updated_at
kogi_points	KP ledger — earn/redeem events	id, entity_id, event_type, amount, balance_after, reference_id, reference_type, timestamp
match_results	MatchEngine output records	id, query_id, subject_a_id, subject_b_id, score, confidence, explanation JSONB, mechanism_type, accepted, timestamp
allocations	AllocEngine decisions	id, mechanism_type, auction_id, winner_id, item_id, clearing_price, scoring_breakdown JSONB, status, executed_at
auction_bids	All bids submitted across all auction types	id, auction_id, bidder_id, amount, bid_type, proxy_max JSONB, status, submitted_at, withdrawn_at
auction_offers	All seller offers	id, listing_id, seller_id, price, offer_type, bundle_items JSONB, auction_config JSONB, status
deal_negotiations	Deal Room negotiation state	id, deal_id, round_number, party_id, action_type (offer/counter/accept/withdraw), terms JSONB, zopa_range JSONB, timestamp
incentive_events	All incentive events (earn, penalty, streak, badge)	id, entity_id, event_type, kp_delta, reputation_delta, badge_id, streak_id, reference_id, timestamp
streaks	Active streak records per entity	id, entity_id, streak_type, current_count, last_activity_at, multiplier, expires_at, status
badges	Badge award records	id, entity_id, badge_type, awarded_at, evidence JSONB
market_rates	PriceEngine rate snapshots	id, category, skill_tag, complexity_band, median_rate, p25_rate, p75_rate, sample_size, computed_at

personalization_profiles	PersonalizationEngine user state	id, user_id, persona, preference_vector JSONB, dominant_categories JSONB, content_config JSONB, segments JSONB, updated_at
--------------------------	----------------------------------	--

12.2 Redis Hot-Path Caches

Cache Key Pattern	TTL	Content
match:user:{id}:opportunities	5 min	Top-20 current opportunity matches for user — invalidated on new listing or profile update
match:listing:{id}:candidates	2 min	Top-50 worker candidates for a listing — invalidated on new worker availability signal
reputation:{entity_id}	10 min	Composite ReputationScore and tier — invalidated on any reputation event
kp:balance:{entity_id}	1 min	Current KP balance — invalidated on any KP earn/redeem event
market_rate:{skill}:{complexity}	1 hour	Median market rate — refreshed hourly by PriceEngine batch
alloc:auction:{id}:book	30s	Live auction state — current bids, clearing price — invalidated on every bid
persona:{user_id}	30 min	Current persona and ContentDeliveryConfig — invalidated on profile update
zopa:{deal_id}	deal lifetime	Computed ZOPA range for active Deal Room — invalidated on each negotiation round

12.3 Technology Stack

Component	Technology	Role	Notes
kogi-game service	Go	Primary REST/gRPC API; orchestrates all mechanism calls	Event producer to game.* Kafka topics
AllocEngine core	Rust (kogi-modules)	High-performance auction clearing, DA algorithm, Shapley approximation	Embedded in kogi-game via FFI

MatchEngine	Scala 3 (kogi-engine)	Multi-dimensional scoring, similarity computation	gRPC port 9100; stateful user profiles
RecommendationEngine	Scala 3 (kogi-engine)	Collaborative + content-based filtering, hybrid scoring	In-memory profile store; Kafka for feedback
PersonalizationEngine	Scala 3 (kogi-engine)	A/B experiments, UCB1 bandit, persona lifecycle	Deterministic assignment via hash
IncentiveEngine	Scala 3 (kogi-engine)	KP accounting, reputation scoring, streak tracking	Writes to game.reputation Kafka topic
PriceEngine	Go service	Market rate estimation; ZOPA computation	Reads from game.bids and game.allocations
PostgreSQL	Primary DB	Persistent state for all game entities	Partitioned by entity_id for large tables
Redis	Cache	Hot-path balance, match, and auction state	LRU eviction; TTL-based invalidation
Kafka	Streaming	All game events streamed for analytics and cross-service coordination	game.* topics; consumed by kogi-engine

13. UX ARCHITECTURE & KEY GAME FLOWS

13.1 Core UI Surfaces for kogi-game

Surface	Game Features Exposed
Home Dashboard	Personalized opportunity feed; reputation tier badge; KP balance; streak status; Oba briefing cards with match and incentive updates
Marketplace Browse	QScore-ranked listings; match-score badges on recommended items; promoted placements (KP/paid); ZOPA hints on listing prices
Exchange Order Book	Real-time bid/offer ladder; auction countdown; proxy bid controls; clearing price trajectory; liquidity depth chart
Deal Room	ZOPA panel (market rate range); offer/counter-offer history; Oba negotiation assistant; milestone scheduler; escrow status
RFP / Proposal Center	Proposal scoring rubric (buyer); proposal competitiveness score (seller); ranked proposal shortlist; one-click accept
Campaign Dashboard	Velocity tracker; success probability gauge; backer list; Oba outreach center; milestone disbursement controls
Reputation Center	Dimension-by-dimension score breakdown; tier progress bar; badge showcase; KP ledger; streak log
Incentive Rewards	KP balance and earn history; available redemptions; streak status; badge library; affiliate commission tracker
Analytics / OKR	PortfolioSignal gauges; health trend charts; risk recommendations; market rate benchmarks; allocation efficiency report
Oba Chat Interface	Natural language access to all mechanism outputs: 'Find me 3 workers for my React project under \$80/hr', 'How competitive is my proposal?'

13.2 End-to-End Game Flows

13.2.1 Worker Wins a Contract via Reverse Auction RFP

32. Buyer posts RFP in kogi-marketplace with scope, budget, and scoring rubric (30% price / 40% skills / 20% reputation / 10% timeline)
33. MatchEngine broadcasts to top-50 worker matches; Oba notifies each in next morning briefing
34. Worker reviews RFP in their Proposal Center; Oba drafts a competitive proposal with market-rate pricing guidance
35. Worker submits proposal; AllocEngine scores it against the buyer's rubric and ranks it against all proposals
36. Buyer reviews ranked shortlist; Oba summarizes top-3 proposals with key trade-offs
37. Buyer accepts top-ranked proposal; Deal Room opens; milestone schedule negotiated
38. Deal agreed; kogi-bank provisions escrow; project appears in worker's kogi-portfolio

39. On deal close: worker earns KP + Delivery Reliability update; buyer earns KP + Financial Integrity confirmation

13.2.2 Asset Creator Builds a Subscription Ecosystem

- 40. User 1 creates Real Estate Investment Playbook as a Portfolio Asset in kogi-portfolio
- 41. Publishes to kogi-marketplace: subscription tier (\$19/mo) + one-time purchase (\$99)
- 42. RecommendationEngine surfaces the playbook to User 2 via content-based filtering (shared real estate interest tags)
- 43. User 2 subscribes; playbook added to their portfolio; User 1 earns KP + subscription revenue in kogi-bank
- 44. User 1 publishes an updated playbook edition; IncentiveEngine applies subscriber discount (20% for 12+ month subscribers)
- 45. Oba notifies User 2 of the new edition with the personalized discount applied
- 46. User 2 purchases; referral-chain tracked if User 2 referred a User 3 with an affiliate link

13.2.3 Cooperative Closes a Sprint and Distributes Patronage

- 47. Cooperative's sprint closes; all task completions recorded in kogi-portfolio
- 48. AnalyticsEngine computes per-member contribution signals (hours, quality scores, deliverables)
- 49. AllocEngine runs Monte Carlo Shapley approximation on contribution vectors
- 50. Oba presents distribution plan to cooperative governance: 'Proposed patronage distribution for Sprint 12: Jordan 34%, Alex 28%, Sam 22%, Taylor 16%'
- 51. Governance vote initiated; members vote via kogi-organizations; quorum reached
- 52. Vote passes; kogi-bank executes batch payroll to all member Payroll Wallets atomically
- 53. All members earn Governance Participation KP; distribution logged in cooperative's ledger

14. OPEN ITEMS & FUTURE CONSIDERATIONS

14.1 Near-Term Open Items

Item	Description
Combinatorial Auction Solver	Combinatorial bids (bundles of resources) require a winner determination algorithm that is NP-hard in general. Implement an approximate solver (LP relaxation + branch-and-bound for small N; greedy approximation for large N) with a bid complexity limit to keep computation tractable.
Shapley Approximation Calibration	Validate Monte Carlo Shapley convergence against exact computation for small cooperatives (≤ 10 members). Establish confidence intervals for approximation accuracy. Define policy for edge cases where contribution vectors are near-uniform.
Multi-Jurisdiction Compliance for Auctions	Real-money auctions with equity instruments may require broker-dealer licensing in certain jurisdictions. Mechanism design must comply with FINRA and SEC Regulation CF for equity crowdfunding above threshold amounts. Legal review required before production deployment of equity campaigns.
KogiPoints Valuation & Anti-Gaming	KP must not become a de facto currency that triggers money transmission licensing. Design redemption mechanics to keep KP as non-transferable, platform-internal credits only. Anti-gaming controls needed for fake review KP farming.
MatchEngine Cold-Start for New Listings	New listings have no interaction history. Current cold-start relies on global popularity signals. Implement category-median profile seeding and seller's historical listing performance as cold-start priors.
Real-Time ZOPA Update	ZOPA computation currently uses a static market rate snapshot. High-volume exchange categories need continuous ZOPA updates as new deals close. Streaming ZOPA update from game.allocations Kafka topic needed.
Federated Game Mechanics	Cross-platform (ShangoOS) game mechanics — how kogi-game interacts with ume's organizational games and qala's solution games — require a defined Federated Mechanism Protocol. Specify the cross-platform incentive alignment rules for jiwe seed loaded environments.
Oba Writeback Protocol for Mechanisms	Oba generates allocation recommendations and draft proposals but requires a defined writeback protocol to submit bids, accept proposals, and execute game actions on behalf of users with pre-approved automation rules — without bypassing PermissionTier checks.

14.2 Phase 4+ Game Mechanics

- Prediction markets — workers and investors bet on project outcomes (will this campaign fund? will this gig deliver on time?); outcomes calibrate RiskEngine priors

- Mechanism tournaments — quarterly platform-wide competitions where participants compete on defined efficiency metrics (fastest gig delivery, highest campaign funding velocity) for prize pool rewards
- On-chain mechanism execution — migrate high-value auction clearing and cooperative distribution to smart contracts for trustless, auditable execution without platform intermediation
- Dynamic fee optimization — replace fixed fee tiers with Myerson-optimal dynamic fees computed per-transaction based on elasticity estimates from PriceEngine
- Cross-platform reputation portability — export Kogi reputation signals to external platforms (Upwork, Fiverr, GitHub) via a verifiable credential standard
- Federated matching — MatchEngine operates across ShangoOS federates: a gig posted in ume is matched by kogi's MatchEngine and fulfilled by a qala solution component
- Simulation environment (oru integration) — run counterfactual simulations of allocation and incentive rule changes in oru before deploying to production, using historical kogi-game data as the simulation substrate
- AI-autonomous market making — Oba-class agents post algorithmic bids and offers to thin markets to maintain price discovery without human participation, acting as algorithmic market makers in underserved categories